

Alpha decay 3

From the previous section we have

$$\tau_{1/2} = \frac{2 \log 2}{\beta} \frac{r_1}{\sqrt{E/m}} e^{2\gamma}$$

We can estimate β from the following half-life data. Rows with $Z = 90$ are uranium and $Z = 88$ are thorium. Recall that Z is atomic number minus two.

$\log \tau_{1/2}$	A	Z	E
6.30	228	90	6.80
14.4	230	90	5.99
21.5	232	90	5.41
29.7	234	90	4.86
34.2	236	90	4.57
39.5	238	90	4.27
7.52	226	88	6.45
17.9	228	88	5.52
28.5	230	88	4.77
40.6	232	88	4.08

Take the log of both sides.

$$\log \tau_{1/2} = -\log \beta + \log \left(2 \log 2 \frac{r_1}{\sqrt{E/m}} \right) + 2\gamma$$

Rearrange as

$$\log \beta = -\log \tau_{1/2} + \log \left(2 \log 2 \frac{r_1}{\sqrt{E/m}} \right) + 2\gamma$$

Let y be the residuals

$$y = -\log \tau_{1/2} + \log \left(2 \log 2 \frac{r_1}{\sqrt{E/m}} \right) + 2\gamma$$

The ordinary least squares estimate of $\log \beta$ is the average of the residuals.

$$\log \beta = \bar{y}$$

From the data we obtain

$$\bar{y} = 6.737$$

Hence

$$\beta = \exp(\bar{y}) = 843$$

The predicted values are

$\log \tau_{1/2}$	Predicted $\log \tau_{1/2}$
6.30	5.02
14.4	13.9
21.5	21.4
29.7	29.9
34.2	34.8
39.5	40.5
7.52	6.52
17.9	17.6
28.5	28.8
40.6	41.7

Eigenmath script